

Software Requirements Specification (SRS)

Coworking Space Conference Room Reservation & Client Usage Tracking System

Document ID:	SRS-CWK-2026-V1.0	Target Release:	Phase 1 MVP
Prepared For:	Engineering & Product Team	Status:	Ready for Development

1. Purpose & Executive Summary

This document details the functional, non-functional, and data tracking requirements for an integrated web portal for coworking space conference room management. The system enables authenticated users to pre-reserve meeting spaces in advance, allows walk-in guests or members to perform on-demand instant bookings, and implements accurate, client-specific time and utilization logging for post-billing and quota controls.

2. User Roles & Access Controls

Client / Member (Tenant) Authenticated corporate client or hot-desk member. Can browse available rooms, view calendar slots, schedule future reservations, execute instant check-in bookings, and view team usage statistics.	Coworking Space Admin Facility manager or desk operator. Can configure rooms, define hourly rates, manage booking overrides, access master calendar views, and export granular usage logs for invoicing.
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3. Core Booking Modes: Comparative Specifications

Feature Aspect	Advance Pre-Reservation	Instant / On-Demand Booking
Trigger & Timing	Scheduled from 2 hours up to 60 days in advance via web portal calendar.	Booked immediately on-site or via mobile web when room is unoccupied.
Slot Allocation	Defined time blocks (e.g., 30-min increments; 09:00–10:30).	Quick start ("Book Now" for 15, 30, 45, or 60 mins) based on immediate vacancy.
Conflict Handling	Strict calendar mutex; prevents double-booking during checkout.	Capped automatically if an advance booking starts within the next hour.
Check-In & Release	Requires digital check-in within 15 minutes of start; else auto-cancelled.	Immediate activation upon button confirmation; no grace period needed.

4. User Authentication & Profile Requirements

- REQ-AUTH-01:** Secure email/password login and corporate Single Sign-On (SSO via Google Workspace / Microsoft Entra ID).
- REQ-AUTH-02:** Multi-tenant client association: Users must be mapped to their parent "Client Account / Company" to pool monthly quota hours.
- REQ-AUTH-03:** Role-based access control (Admin, Client Lead/Manager, Standard Member).

5. Functional Requirements: Reservation & Instant Booking

Req ID	Module	Requirement Description	Priority
REQ-RES-01	Room Discovery	Search by capacity (seats), amenities (whiteboard, AV display, video conference bar), and date/time window.	MUST
REQ-RES-02	Interactive Grid	Visual timeline grid showing room occupancy, cleaning buffer periods (10 min auto-buffer), and available slots.	MUST
REQ-RES-03	Instant Booking	"Book Now" quick-action on dashboard showing only currently unoccupied rooms. Single-click reservation up to the next booked slot.	MUST
REQ-RES-04	Session Extension	In-room instant extension request: If no following reservation exists, allow 15-min or 30-min incremental extensions.	SHOULD
REQ-RES-05	Cancellations	Self-serve cancellation up to 1 hour before scheduled start without quota penalty. Dynamic penalty calculation if canceled late.	MUST
REQ-RES-06	Reminders	Automated email/SMS notification sent 15 minutes prior to booking start time with one-touch check-in link.	SHOULD

6. Client Usage Time Tracking & Accounting (Critical Focus)

Accurate time tracking per corporate client is the primary business metric for billing, tier monitoring, and dispute resolution.

Core Principle: Every session must track two separate timestamps: *Scheduled Duration* and *Actual Consumed Duration* (tracked via Check-in and Check-out / Tap-out actions).

Log ID	Tracking Field	Specification & Validation Logic	Storage Type
REQ-TRK-01	Client Identification	Company ID, Department code, and Individual Booker ID permanently tagged to record.	UUID / Indexed
REQ-TRK-02	Time Logs	Scheduled Start/End vs. Actual Check-In and Check-Out timestamps stored in UTC with local TZ display.	Timestamp with TZ
REQ-TRK-03	Overstay Detection	If user does not check out when scheduled time lapses, flag session as "Overstayed" and log incremental billable units.	Boolean + Minutes
REQ-TRK-04	Quota Deductions	Real-time decrement against client's pre-paid monthly allocation (e.g., 20 free hours/month). Alerts when balance < 2 hours.	Float (Hours/Mins)

6.1 Usage Reporting & Export Capabilities

- **Client Dashboard:** Monthly burn chart showing hours consumed vs. contractual monthly quota.
- **Admin Analytics:** Utilization rates by room (e.g., Boardroom A: 72% occupancy) and client breakdown tables.
- **Data Export:** One-click CSV and JSON reports structured for direct ingestion into accounting software (QuickBooks / Xero).

7. System Architecture & Data Model Overview

The following conceptual data schema must be implemented to support atomic booking transactions and accurate client usage logging:

Entity	Key Attributes	Relationship & Constraints
ClientCompany	id, name, billing_tier, monthly_quota_hrs, status	One-to-many with Users; One-to-many with Bookings.
ConferenceRoom	id, name, capacity, equipment_list, hourly_rate, is_active	Has many Bookings and maintenance schedules.
Reservation	id, client_id, user_id, room_id, type [PRE INSTANT], status	Status: PENDING, CONFIRMED, CHECKED_IN, COMPLETED, CANCELLED.
UsageLog	id, reservation_id, client_id, start_time, end_time, billable_mins	Immutable audit log generated upon room checkout.

8. Non-Functional & Engineering Constraints

Concurrency & Race Conditions

Database transaction isolation (Serializable or Row-Level Locks) must be enforced during the booking step to guarantee that two clients booking the same slot simultaneously cannot create conflicting records.

Performance & Latency

The Instant Booking action must complete in < 1.2 seconds from user tap to confirmation response, accommodating walk-up tablet or mobile devices.

Mobile Responsiveness

Interface must be fully optimized for handheld smartphones (members in transit) and 10-inch room display tablets mounted outside meeting doors.

Auditability & Security

All time logs, extensions, and manual admin adjustments must produce append-only audit trail entries with actor identification and timestamps.

9. Acceptance Criteria for Development Team (Sign-Off)

- AC-01:** A registered user can view open slots and confirm a reservation 14 days out within 3 clicks.
- AC-02:** An instant booking option is visible on empty rooms, immediately locking the room in real-time across all user dashboards.
- AC-03:** Usage time calculation accurately attributes billable minutes to the correct client profile, including auto-checkout when sessions end.
- AC-04:** Admins can export a client-itemized usage report filtered by custom date ranges in CSV format with 100% time reconciliation.